

Education

University of Florida, Ph.D. candidate in the ECE Department	Aug. 2022 – May. 2027
Southern University of Science and Technology, Bachelor of Engineering	Sep. 2018 – Jun. 2022

Professional Summary

Ph.D. researcher in **AI agents and multimodal learning**, with industry experience at **ByteDance**. Built and launched an agent system supporting **100+ workflows and 40+ users**. Experienced in VLM fine-tuning, agent tooling, and evaluation, with specialized expertise in AI privacy and security.

Professional Experience

- Machine Learning Engineer Intern**, ByteDance Inc. E-commerce Group – San Jose, CA May 2026 – Aug. 2026
- Built and launched a **self-improving agent system** with **MCP integration** for e-commerce user-risk detection; scaled to **100+ workflows and 40+ users** across internal teams.
 - Developed a **CLI** for a GUI-based workflow platform, enabling the agent to retrieve **workflows and execution traces** for automated diagnosis and optimization.
 - Improved risk-detection recall from **63.1% to 94.3%** while reducing the **false-positive rate by 87%** through trace-driven refinement of workflow prompts, code, and data-processing logic.
 - Designed **workflow validation and change controls**: Python schema/tool checks and protected-node constraints bounded edits, while **LLM-as-a-Judge** assessed groundedness and reasoning quality for candidate acceptance.
 - Implemented **automated fault recovery** with retries, fallbacks, and downstream validation to distinguish unavailable data from valid results, reducing execution failures from **80% to near zero** on targeted high-failure workflows.
 - Integrated the agent into an **award-winning human-in-the-loop (HITL) initiative**, which ranked **5th among 40 projects** in a **department-wide competition** within ByteDance’s Governance and Experience (GNE) organization.
- Research Assistant**, University of Florida – Gainesville, FL Aug. 2022 – Present
- Developed **Python/PyTorch** systems for **VLM fine-tuning**, multimodal privacy evaluation, adversarial robustness, federated learning security, and ML-based vulnerability analysis.
 - Published peer-reviewed research in **USENIX and IEEE venues**; built reproducible pipelines for model training, fine-tuning, and security evaluation.

Projects

PrivacyProbe: Automated Privacy Red Teaming for Conversational AI

- Built a **privacy red-teaming framework** to evaluate sensitive-attribute elicitation across four service personas; used **structured conversational memory** to track inference evidence and guide adaptive questioning.
- Evaluated **30 synthetic profiles, 19 attributes, and 15-turn conversations**; achieved **75.3% macro-averaged inference accuracy** in the life-coach simulation, exceeding non-adversarial and passive baselines by **29.2 and 49.5 percentage points**; ablated memory and target selection to assess their contributions.

Multimodal PCB Analysis with VLM Fine-Tuning and RAG

- Fine-tuned **Qwen2.5-VL** with **LoRA** using **LLaMA-Factory** and **PEFT** for **multimodal supervised fine-tuning (SFT)**; configured FP16 mixed precision and gradient accumulation on a single GPU.
- Curated a **multi-image, multi-turn instruction dataset** spanning **38 PCB designs and 254 images**, covering schematics, component placement, and PCB layers for electromagnetic interference (EMI) analysis.
- Built **text RAG with LangChain, FAISS, and Sentence Transformers**, combining retrieved layout guidance with device-metadata lookup in a three-stage VLM reasoning pipeline; implemented a separate **Gemini API** pipeline for EMI report generation.

VLM Security: Adversarial Detection and Jailbreak Defense

- Implemented a **quantization-based defense** against adversarial VLM jailbreaks, achieving a **95% defense success rate**; developed a **ViT-ResNet meta-learning detector** for adaptation to emerging adversarial inputs.
- Optimized robustness through **cross-modal gradient alignment** between visual and text embeddings and evaluated transferability across multiple open-source VLMs.

Apple Vision Pro: Confirmed Privacy Vulnerability

- Identified an **Apple-confirmed vulnerability** and validated a remote keystroke inference attack on **Apple Vision Pro**.
- Implemented an ML attack pipeline using **3D gaze estimation, typing-session detection, and top- K prediction** to infer keystrokes from avatar gaze observations.

ShareGuard: Context-Aware Multimodal Privacy Review Engine

- Built a **four-stage privacy review pipeline** for image sharing; separated sensitive-content extraction and audience-boundary inference from recipient matching to reduce recipient-induced decision bias.
- Benchmarked **five VLMs on 160 cases** against text-only, image-only, and rule-lookup baselines; improved **GPT-4.1 sharing-decision accuracy from 75.0% to 91.2%**.

Federated Learning Security: Targeted Poisoning Evaluation

- Developed a **targeted model-poisoning attack** against federated learning systems, reducing target-class accuracy to **35%** while limiting changes in non-target-class performance.
- Used **KL divergence** to compare model-weight distributions and guide attack stealth; evaluated evasion of **Krum** and **k -out-of- n selection** defenses.

Technical Skills

Languages: Python, SQL, Bash/Shell, MATLAB

Model Training: PyTorch, Hugging Face Transformers, LLaMA-Factory, PEFT, LoRA, Multimodal SFT

AI Security: Privacy Red Teaming, Adversarial Robustness, Jailbreak Defense, Model Poisoning

Agents/Tools: AI Agents, MCP, CLI Development, Structured Memory

Evaluation: LLM-as-a-Judge, Schema Validation, Human-in-the-Loop

Retrieval/APIs: Text RAG, LangChain, FAISS, Sentence Transformers, Gemini API

Infrastructure: AWS, Linux, Git

Selected Publications

USENIX 2024. Hanqiu W. et al. “GAZEexploit: Remote Keystroke Inference Attack by Gaze Estimation from Avatar Views in VR/MR Devices.”

IEEE ICC 2025. Siqi Dai et al. “Guided by Noise: Vulnerable Poisoning Attack to Differentially Private Federated Learning.”

ICRA 2024. Zero-shot Safety Prediction for Autonomous Robots with Foundation World Models.

PCIM 2026. EMI Diagnosis for Power Converter PCB Layouts based on a Reasoning Aligned Vision Language Model.